

An open tank system is shown in Figure Q1. The system consists of one inlet liquid flow (F_i) and one outlet flow through a flow control valve (F_o).

Where,

h - liquid level in the tank (m)

R - valve constant

A - cross-sectional area of the tank (m^2)

Assume that $A=2 m^2$, $R=10$ and initial condition deviation variable, $h'(t=0)=0$.

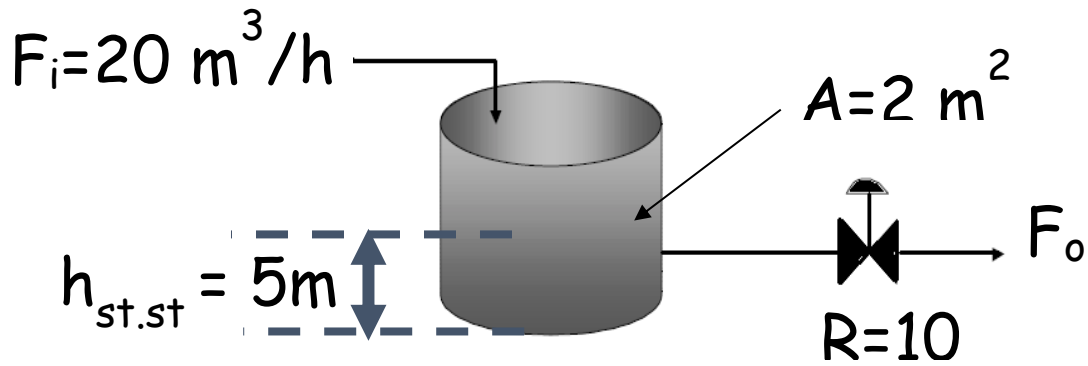


Figure Q1

Plot the $h(t)$ function, increasing the inlet liquid flow (F_i) to $30 \text{ m}^3/\text{h}$.